Flooding the filter: a game-theoretic model of AI, submission pressure, and quality collapse

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Significance. Artificial intelligence makes it dramatically easier to produce a manuscript but not to verify one. We show that this asymmetry can silently erode peer review until the system crosses a critical transition into a self-reinforcing low-quality state. Using a game-theoretic model of author strategies under AI adoption, we find that gradual increases in AI-assisted submission can trigger an abrupt collapse in publication quality, and that recovery from collapse requires far more effort than prevention. Expanding the number of publication slots without investing in verification makes the problem worse. These results imply that institutional reform should prioritise verification infrastructure—screening quality, review capacity, and submission friction—over simply publishing more. The window for preventive action is narrower than current quality metrics suggest.

Abstract

Generative AI has made it far cheaper to produce a manuscript while the capacity to verify one has grown slowly. We model this asymmetry using evolutionary game theory: AI adoption simultaneously raises submission volume, lowers drafting cost, degrades screening reliability, and only weakly increases review capacity. The net effect is to raise the expected payoff from submitting low-quality work. We find that this payoff rises monotonically with AI adoption, so that a system initially sustaining honest submission norms can undergo a critical transition into a low-quality equilibrium. Once collapsed, recovery is asymmetric: the level of AI adoption at which quality collapses is strictly lower than the level at which it recovers, trapping the system in a degraded state. Expanding publication capacity without scaling verification accelerates the collapse—more slots raise the prize in a game whose entry barrier has already been eroded. Agent-based simulations confirm these dynamics under finite-population noise and heterogeneous adoption. Our analysis suggests that effective institutional reform must target verification directly—strengthening screening quality, expanding review capacity, and restoring submission friction—rather than simply adding publication slots.

1 Introduction

Generative AI is reshaping the production of scientific knowledge at a pace that outstrips the capacity of existing institutions to absorb it. By September 2024, more than one-fifth of computer-science preprints on arXiv showed measurable evidence of large-language-model modification (Liang et al., 2025), and at least 13.5% of all biomedical abstracts published that year contained AI-assisted text (Kobak et al., 2025). Monthly submissions to arXiv reached a record 24,226 in October 2024, roughly 50% higher than in late 2022 (arXiv, 2024). While production has surged, verification has not kept pace: the rate at which researchers accept reviewer invitations has declined steadily over two decades (Meyerson et al., 2025), and between 6.5% and 16.9% of conference peer reviews now contain substantial AI-generated text (Liang et al., 2024).

This asymmetry between production and verification is not unique to publishing. Historically, technologies that accelerate the creation of knowledge have outpaced the institutions designed to evaluate it, and sustained progress has depended on building institutional bridges between production and verification (Mokyr, 2002; Walker, 2026). AI intensifies this pattern: it automates precisely the tasks that are cheapest to verify—drafting, formatting, stylistic polish—while leaving the expensive tasks—checking data, replicating analyses, evaluating genuine novelty—largely untouched (Walker, 2026; Checco et al., 2021).

At the population level, AI-adopting scientists publish substantially more but follow narrower research trajectories, with distinct topics contracting and follow-on engagement between researchers declining (Hao et al., 2026). The question is not whether individual researchers behave ethically, but whether the *system* as a whole can absorb an asymmetric shock to production and verification.

While the conventional response to AI in science focuses on disclosure requirements, detection tools, and editorial guidelines, these measures address individual conduct and leave untouched a structural problem: even if every author is honest, reviewers can scrutinise only a finite number of manuscripts carefully per cycle. Peer review is a shared resource under increasing strain: the global system already consumed over 100 million hours of researcher time each year before AI entered the picture (Kovanis et al., 2016; Aczél et al., 2021), and screening quality declines even when reviewers act in good faith, simply because competing demands leave too little time per paper (Bianchi et al., 2018). When submission volume rises faster than review capacity, screening degrades—not because of misconduct, but because of congestion (Gross and Bergstrom, 2019). We focus on this feedback between volume and screening quality as the central mechanism.

The consequences of this feedback are not merely individual but collective. When the review filter is imprecise, submitting everything regardless of quality can become individually rational (Thurner and Hanel, 2011), and prestige-maximising incentives can reward strategic over honest behaviour (Zollman et al., 2024). The honest-submission norm is fragile: it collapses when the cost of mimicking quality falls even slightly (Tiokhin et al., 2021)—precisely the shift that AI-assisted writing produces. "Publish or perish" incentives already act as a selection force that spreads high-volume, low-quality strategies through academic communities (Smaldino and McElreath, 2016; Ioannidis, 2005). Because the outcome depends not on any single author's choice but on how submission strategies spread through a population of researchers responding to shared incentives, we frame the problem using evolutionary dynamics (Weibull, 1995; Hofbauer and Sigmund, 1998). Zhong et al. (2026) show that when AI adoption spreads through a population, individually rational choices can erode the collective capacity for exploration, and that group-level institutional structures are needed to counteract this drift.

While existing literatures each address part of this problem, none unifies them. Models of strategic submission do not treat AI as a structural shock that shifts multiple inputs at once (Thurner and Hanel, 2011; Zollman et al., 2024; Tiokhin et al., 2021). The empirical AI-in-science literature documents faster production, slower review growth, and higher false-positive risk, but does not connect these to a model with clear predictions about the long-run state of the system (Noy and Zhang, 2023; Liang et al., 2025; Latona et al., 2024). The institutional design literature asks how gatekeeping should work (Ellison, 2002; Muller-Itten, 2022; Cotton, 2013) but does not ask how its proposals must change when drafting cost approaches zero and the screening signal degrades continuously with adoption. We close these gaps by connecting all four channels of AI influence—rising volume, falling cost, degraded screening, and lagging review capacity—to the long-run state of the publishing system, characterising conditions under which quality collapses suddenly and is hard to restore, and deriving which institutional levers can prevent or reverse that shift.

2 Results

2.1 AI as an Asymmetric Shock to the Publication System

We model a population of N authors submitting to a single journal over repeated periods. Each manuscript has binary quality: good with probability α, and bad with probability $1 - \alpha$. Authors choose between two strategies: *selective submission*, in which an author submits only genuine work, and *universal submission*, in which an author submits regardless of quality, hoping that imperfect screening allows weak papers through. We write $x \in [0, 1]$ for the share of authors playing the selective strategy. Under replicator dynamics, intermediate submission strategies are evolutionarily unstable in the mean-field limit, reducing the effective strategy space to these two options (Weibull, 1995; Hofbauer and Sigmund, 1998).

Generative AI affects the publication system through four channels simultaneously, captured by a single adoption parameter $a \in [0, 1]$ representing the field-level intensity of AI use. First, AI raises total

submission volume by lowering the marginal cost of drafting and preparing a manuscript; randomised experiments find that large-language-model access cuts professional writing time by roughly 40% (Noy and Zhang, 2023; Stokel-Walker and Van Noorden, 2023). Second, AI directly lowers the cost of each submission attempt, asymmetrically benefiting authors who submit indiscriminately because the same per-paper reduction compounds across a larger submission stream. Third, AI-generated text is harder to distinguish from genuine contributions on surface features (Stokel-Walker and Van Noorden, 2023), AI-assisted reviews award systematically higher scores (Latona et al., 2024; Liang et al., 2024), and automated detection remains unreliable (Liang et al., 2023)—each mechanism raises the probability λ that a bad paper passes review. Fourth, AI can assist reviewers by summarising papers and flagging inconsistencies, raising review capacity—but deep verification remains human-intensive and policy-restricted (Naddaf, 2025), so this channel grows more slowly than the others. The structural asymmetry is captured by the condition:

$$\gamma > \delta \geq 0, \tag{1}$$

where γ governs how strongly AI raises submission volume and δ governs how strongly it raises review capacity. This is not an axiom but a testable empirical statement: submission volume grows faster with AI adoption than review capacity does. The functional forms for each channel are given in the Materials and Methods; Figure S3 in the Supplementary Material summarises all four channels and their functional forms.

The publication process has two distinct capacity constraints. The first is *verification capacity* K_{rev}: the number of papers that can receive careful scrutiny per period. When submission volume exceeds review capacity, scrutiny per paper falls and effective screening degrades—not because reviewers are careless, but because there are too many papers for the available expert time (Bianchi et al., 2018; Kovanis et al., 2016). The second is *allocation capacity* K: the number of publication slots the journal can fill per period. When the pool of papers that pass review exceeds K, the journal must ration: each paper faces a publication lottery, and the probability of acceptance falls below one.

The strategic difference between selective and universal submission reduces to the net value of submitting a weak paper:

$$\Pi_{bad}(x, a) = r \cdot \bar{\rho}(x, a) \cdot \pi_0(x, a) - c(a), \tag{2}$$

where r is the publication reward, $\bar{\rho}$ is the probability that a paper passing review is ultimately published (the rationing probability), π_0 is the probability that a bad paper passes review, and $c(a)$ is the submission cost, which falls with AI adoption. When $\Pi_{bad} > 0$, submitting weak work is individually profitable and the universal strategy spreads; when $\Pi_{bad} < 0$, the selective strategy dominates. The population dynamics follow the standard replicator equation:

$$\dot{x} = -x(1 - x)(1 - \alpha)\, \Pi_{bad}(x, a). \tag{3}$$

Two forces compete as the share of universal submitters grows. Rising volume degrades screening (the bad-paper pass rate π_0 rises), making it more attractive to submit weak papers. But rising volume also saturates the publication queue (the rationing probability $\bar{\rho}$ falls), making the lottery harder to win. While the screening-collapse force pushes towards universal submission, the rationing force can counteract it: when slot scarcity is extreme, the lottery deters indiscriminate submission even after screening has failed—a mechanism we call *defence by congestion*.

The signs of Π_{bad} at the two boundary states determine the long-run outcome. When the net payoff from weak submissions is negative everywhere, the system converges to *Global Honesty* ($x = 1$): all authors submit only genuine work. When it is positive everywhere, the system collapses to *Global Corruption* ($x = 0$): everyone submits indiscriminately. When the net payoff from weak submissions is negative at the honest boundary but positive at the corrupt boundary, both extremes are stable and an unstable collapse threshold separates their basins of attraction—this is *Bistability*, where history determines the outcome and recovery from corruption requires a large intervention. A key structural observation is that the two boundaries are controlled by different parameters: the honest boundary ($x = 1$) is secured by keeping λ low, while the corrupt boundary ($x = 0$) is held by keeping K tight (Figure S1 in the Supplementary Material illustrates the bistable case; Figure S2 traces the equilibrium trajectory under two parameter sets).

The four AI channels act jointly on Π_{bad}. At the honest boundary, where only good papers are submitted and review capacity is not the binding constraint, the net payoff from weak submissions simplifies: a

bad paper passes at rate $\lambda(a)$ and costs $c(a)$ to submit. The collapse threshold—the level of AI adoption above which the honest equilibrium is no longer self-sustaining—has a closed form:

$$a^* = \frac{c_0 - r\lambda_0}{r\mu + c_0\phi}, \tag{4}$$

where c_0 is the baseline submission cost, λ_0 the baseline false-positive rate, μ the rate at which AI degrades screening, and ϕ the rate at which AI erodes submission friction. For a typical journal prone to bistability (Set A parameters: $K_{rev} = 300$, $K = 400$, $r = 100$, $c_0 = 25$, $\lambda_0 = 0.1$), this threshold is $a^* = 0.50$. Above a^*, no institutional configuration sustains Global Honesty without additional intervention.

Not every AI adoption trajectory leads to quality collapse. If review capacity scales nearly as fast as submission volume ($\delta \approx \gamma$) and AI tools do not degrade screening reliability ($\mu \approx 0$), the cost reduction alone is typically insufficient to cross the collapse threshold. The critical implication is that the institutional response determines which scenario the system occupies: while AI adoption is largely exogenous, the verification gap ($\gamma - \delta$) and screening vulnerability (μ) are policy choices.

2.2 AI Shifts Regime Boundaries

Figure 1 presents the core comparative-statics result. The left panel shows the baseline ($a = 0$) regime map across the space of review capacity (K_{rev}) and publication slots (K). The right panel shows the same map under high AI adoption ($a = 0.8$). The comparison is unambiguous: the Global Honesty region shrinks substantially, the Bistability and Global Corruption zones expand, and the Stable Coexistence region contracts. A journal safely in the honest zone at baseline may find itself in bistability or corruption at $a = 0.8$ without any change in its own institutional parameters.

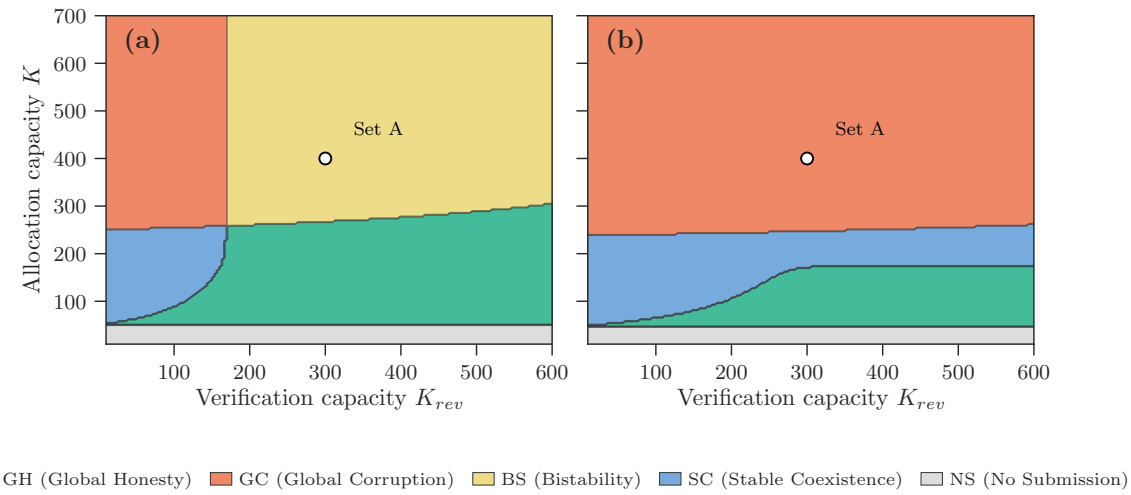

Figure 1: **AI adoption compresses the region of parameter space where honest submission norms are self-sustaining.** Each panel maps the long-run outcome across review capacity and publication slot capacity. Left: pre-AI baseline. Right: high AI adoption ($a = 0.8$). The green region (honest norms sustainable) shrinks markedly; the yellow (bistable, history-dependent) and red (universal corruption) regions expand. A journal represented by the star sits safely in the honest zone at baseline but approaches the boundary of bistability under AI adoption—without any change in its own policies. (Parameters: $\gamma = 0.5$, $\delta = 0.1$, $\phi = 0.4$, $\mu = 0.2$; see Materials and Methods for full calibration.)

The magnitude of this shift is quantifiable. At $a = 0.8$, submission volume has risen by 40% while review capacity has risen by only 8%. A journal with $K_{rev} = 300$ now has the effective screening power of a journal with $K_{rev} \approx 231$ at baseline—a loss of nearly one-quarter of its screening capacity with no change in institutional design.

Expanding publication slots seems like an obvious response to submission pressure, but this response

backfires when verification does not scale proportionally. Increasing K raises the rationing probability $\bar{\rho}$, which raises the expected payoff from submitting a weak paper. AI amplifies this effect: when submission cost is already low and screening already degraded, even a modest improvement in the publication lottery is enough to push the system from honesty into bistability. More slots increase the prize in a game whose entry barrier has already been eroded—a *slot expansion paradox* that AI adoption amplifies rather than attenuates.

2.3 Critical Transitions and Asymmetric Recovery

Figure 2 makes the asymmetric recovery structure explicit, plotting the long-run honest-author share against AI adoption under two field-pressure levels.

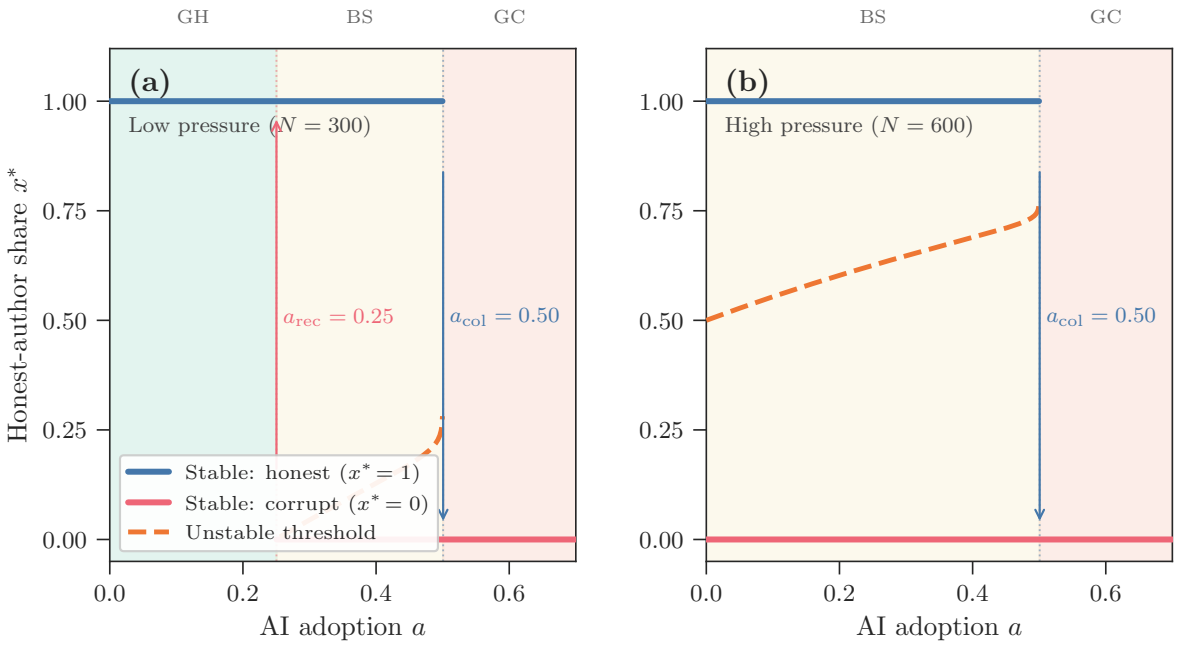

Figure 2: **Recovery from collapse requires far more effort than prevention: the system exhibits asymmetric recovery.** Each panel plots the long-run honest-author share against AI adoption. **(a)** A low-pressure field starts honest and collapses at $a \approx 0.50$; recovery requires reducing adoption to $a \approx 0.25$—a recovery gap of 25 percentage points. The dashed curve marks the unstable collapse threshold separating the two basins in the bistable corridor. **(b)** A high-pressure field starts in bistability before any AI adoption; the corrupt equilibrium persists for all adoption levels, so reducing AI adoption alone cannot restore honest norms. (Parameters: Set A; (a) $N = 300$, (b) $N = 600$.)

In panel (a), the system occupies Global Honesty at baseline. As AI adoption rises, the corrupt equilibrium first becomes self-sustaining at $a \approx 0.25$, opening a bistable corridor; the honest equilibrium then loses stability at $a = 0.50$, pushing the system into Global Corruption. While the forward path (increasing a) traces the honest branch to the collapse point, the backward path (decreasing a) recovers only when adoption falls 25 percentage points below the collapse threshold. This recovery gap measures the depth of the trap: a field that has collapsed must substantially reduce AI-linked submission pressure, not merely stop increasing it.

Panel (b) shows a qualitatively different situation. At higher field pressure ($N = 600$), the corrupt equilibrium is already self-sustaining before any AI adoption. There is no recovery threshold in the admissible range: reducing AI adoption alone cannot restore honesty. Escaping the corrupt basin requires an institutional intervention that raises review capacity, reduces the false-positive rate, or both—a shift in the parameter configuration, not merely a reversal of adoption. The contrast between panels captures the core asymmetry of the recovery problem: while low-pressure fields retain a restoration option, high-pressure fields require coordinated institutional action regardless of what happens to AI adoption.

2.4 Silent Erosion of Stability

The system can appear healthy while structural fragility accumulates. Figure 3 makes this concrete.

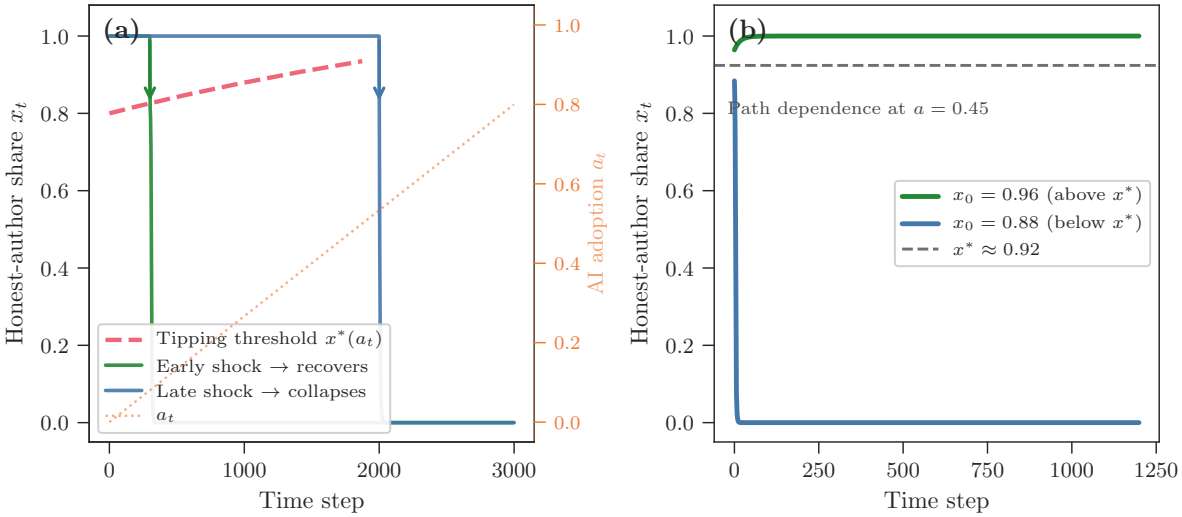

Figure 3: **An identical shock applied at different stages of AI adoption produces opposite outcomes. (a)** The collapse threshold (red dashed) rises silently as AI adoption increases. A fixed-size shock applied early, when the safety margin is wide, produces full recovery (green trajectory); the same shock applied later, when the margin has narrowed, triggers irreversible collapse (blue trajectory). The two trajectories are indistinguishable before the shock. **(b)** Two populations differing by only 8 percentage points in initial honesty diverge to opposite long-run states, confirming that the collapse threshold is a genuine basin boundary. (Parameters: Set A, $\gamma = 0.5$, $\delta = 0.1$, $\phi = 0.4$, $\mu = 0.2$.)

Under Set A parameters, the collapse threshold starts at approximately 80% honest authors and climbs monotonically towards 100% as AI adoption deepens. An identical shock—representing, for example, a coordinated wave of low-quality submissions of the kind documented at major publishers during paper-mill incidents (Bik et al., 2016)—produces full recovery when applied early (the safety margin is wide) but triggers irreversible collapse when applied later (the margin has narrowed). The practical implication is striking: published quality can remain high while the distance to collapse shrinks silently, giving no reliable warning before a critical transition (Scheffer, 2009).

Figure 4 maps this dynamic across the plane of author population size and AI adoption.

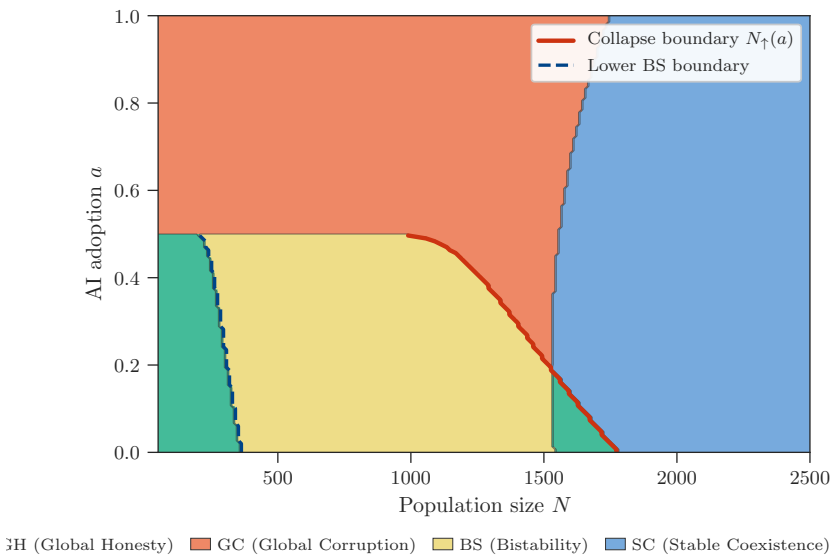

Figure 4: **AI adoption lowers the population threshold at which honest norms collapse.** The red curve marks the collapse boundary: to its right, the system transitions out of the honest zone. As AI adoption increases, the collapse threshold falls monotonically; at $a \gtrsim 0.5$, the honest zone disappears entirely. Recovery from this state requires not merely a reduction in population pressure but a qualitative change in institutional parameters. (Parameters: Set A, $K_{rev} = 300$, $K = 400$.)

At $a = 0$, a population of roughly 1,800 authors is needed to push the system out of the stable-honesty zone; at $a = 0.5$, this threshold has fallen to roughly half. At $a \gtrsim 0.5$, the bistable zone disappears entirely: there is no longer any starting position from which honest norms are self-sustaining. AI adoption transforms the challenge from maintenance to repair.

Among the four AI channels, rising λ is structurally the most dangerous. The honest-boundary condition requires $\lambda < c(a)/r$; under baseline parameters, this requires $\lambda < 0.25$. A slope of $\mu = 0.2$ over full adoption pushes λ from 0.10 to 0.30, crossing this threshold and making Global Honesty structurally impossible regardless of review capacity. No investment in review staffing can restore honest norms once $\lambda \geq c/r$, because even a perfectly reviewed bad paper passes at rate λ and yields a positive expected payoff.

The model identifies observable early warning signals that precede collapse: rising submission-to-review ratios, lengthening turnaround times, increasing desk-rejection rates, and acceptance rates rising faster than improvements in underlying research quality. In fields where AI adoption is high, these signals should appear before any observable quality decline, because the safety margin narrows while published quality appears unchanged.

Agent-based simulations confirm that the three main results—gradual onset of fragility, abrupt collapse, and severe path dependence—survive finite-population noise and heterogeneous AI adoption diffusion (Appendices B and C of the Supplementary Material). A safety margin of even 2 percentage points above the collapse threshold prevents collapse across all community sizes tested ($N \in \{200, 500, 1{,}000\}$), while the absence of a margin exposes roughly half of all runs to quality decline regardless of population size. Robustness analysis across the $(\delta/\gamma, \mu)$ plane reveals that the false-positive slope μ is the binding constraint: even if review capacity scales perfectly with submission volume, the system remains in Global Corruption whenever $\mu > 0.09$.

2.5 Intervention Must Target Verification

Three structural results follow from the analysis (formal statements and proofs in Appendix A of the Supplementary Material). First, AI adoption monotonically destabilises honest submission norms: the net payoff from weak submissions Π_{bad} is strictly increasing in a under the asymmetry condition, and the Global Honesty region in parameter space shrinks as adoption deepens (*monotone destabilisation*).

Second, expanding publication slots raises Π_{bad} at every composition of the author population, and the marginal destabilising effect of an additional slot is strictly larger at higher a (*slot expansion paradox*). Third, the slot constraint required to deter corruption via congestion is strictly tighter at higher a, so journals relying on passive slot scarcity for protection may need to actively reduce K as adoption deepens (*erosion of congestion defence*). Among the primary policy levers, reducing λ and raising K_{rev} are unconditionally beneficial; raising submission friction c is effective but imposes symmetric costs on all authors; and expanding K alone is the unique intervention that can make the system strictly worse.

Figure 5 illustrates the joint effect of institutional interventions.

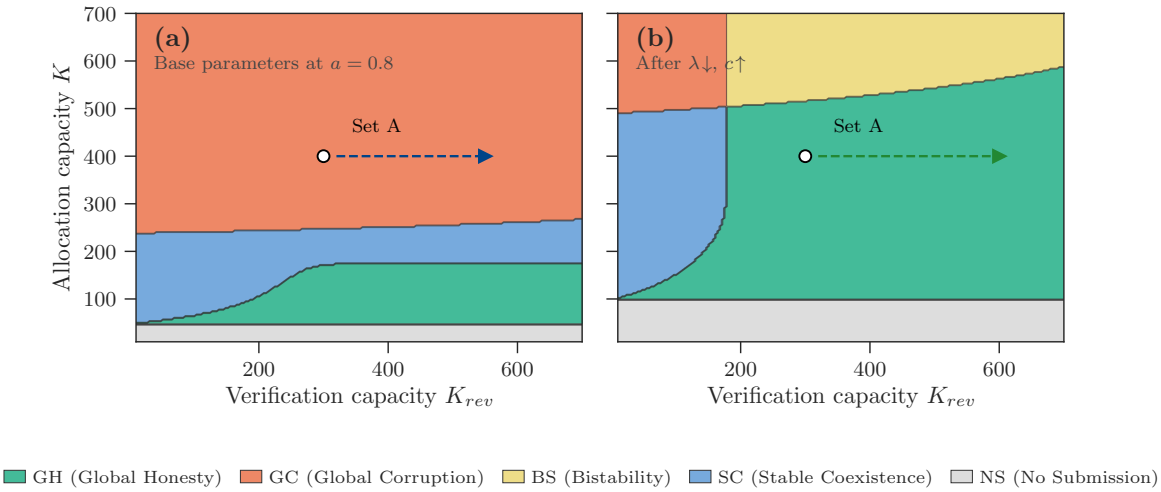

Figure 5: **Verification-focused intervention restores honest norms where capacity expansion alone cannot.** *Left:* Under high AI adoption with baseline parameters, corruption dominates most of the parameter space; expanding review capacity alone (navy arrow) is insufficient because the screening signal is too degraded. *Right:* After strengthening screening quality and restoring submission friction, the corruption zone is eliminated; combining this with expanded review capacity (green arrow) restores the honest zone. The lesson is that screening-quality improvement and capacity expansion are complements, not substitutes. (Parameters: $a = 0.8$, $\gamma = 0.5$, $\delta = 0.1$; Set A baseline.)

The left panel shows that under high AI adoption with baseline parameters, Global Corruption dominates most of the parameter space. Raising review capacity alone leaves the system in corruption because the false-positive rate ($\lambda = 0.26$) exceeds the cost-to-reward ratio ($c/r = 0.17$): even perfectly staffed review cannot restore honesty when the screening signal itself is degraded. The right panel applies screening-quality improvement and submission-friction restoration jointly; the corruption zone is eliminated entirely, and combining this with expanded review capacity restores Global Honesty. The two levers—screening quality (λ down) and review capacity (K_{rev} up)—must be deployed together; neither alone is sufficient.

3 Discussion

The production–verification gap as a recurring pattern. The core vulnerability is the gap $\gamma - \delta > 0$, not AI adoption per se. If review capacity scaled at the same rate as submission volume and AI tools improved screening reliability rather than degrading it, the honest zone would expand under rising adoption, not contract. Noy and Zhang (2023) document genuine productivity gains from AI assistance; the issue is not that AI exists but that the gap between production and verification growth rates exists. Acemoglu and Restrepo (2019) provide a useful framing: automation raises productivity in tasks that are routine and codifiable while increasing demand for tasks requiring contextual judgement. While text generation is largely the former, verifying the validity of a novel empirical claim is largely the latter. This tension between accelerating production and lagging verification is not new—Mokyr (2002) documents it recurring across several centuries of scientific institution-building—but AI intensifies it by operating

on all four channels simultaneously.

Implications for institutional design. The model reduces the institutional problem to a single condition: keep $\Pi_{bad}(x, a)$ negative. Five principles follow, ordered by the tightness of the constraint each relieves.

The first and most critical is to strengthen the screening signal itself. The honest-boundary condition requires $\lambda < c(a)/r$; this is the binding constraint, and no amount of review-capacity expansion or slot restriction restores honesty while λ exceeds this threshold. Mandatory data, code, and materials sharing makes fabrication costlier and enables post-publication verification (Munafò et al., 2017; Nosek et al., 2015). Structured reporting checklists—CONSORT for randomised trials (Moher et al., 2009), PRISMA for systematic reviews—force authors to document methods in ways that make gaps visible. Registered Reports eliminate outcome-selective reporting by committing authors to their design before results are known (Nosek et al., 2018).

The second is to scale verification capacity. Higher review capacity raises the screening intensity, reducing the bad-paper pass rate and lowering Π_{bad} at the corrupt boundary. Practical mechanisms include structured reviewer panels with recognition credits (Aczél et al., 2021), AI-assisted format pre-screening that frees expert time for substantive judgement (Checco et al., 2021), and co-review programmes that train junior reviewers alongside senior ones. Crucially, review assistance tools must not themselves raise λ: format checks are appropriate, but automated quality scoring is not (Naddaf, 2025).

The third is to add smart friction. As AI lowers $c(a)$, the gap $c(a) - r\lambda(a)$ closes and eventually inverts. Submission tokens or annual per-journal quotas raise the effective cost of marginal submissions without requiring any detection of AI use (Cotton, 2013). An equity caveat applies: blunt friction raises costs for all authors, including those at under-resourced institutions; targeted exemptions—token grants, preference for low-infrastructure verification tools such as checklists over high-infrastructure requirements—can preserve incentive effects while redistributing the cost.

The fourth is to decouple dissemination from certification. The current architecture conflates two functions: disseminating findings quickly and certifying quality carefully. When certification degrades under submission pressure, both fail together. Preprint servers combined with overlay journals allow dissemination to remain immediate while certification proceeds at the pace required for genuine scrutiny (Ginsparg, 2011). The slot expansion paradox applies specifically to certified publication slots, not to preprint availability.

The fifth, operating on a longer horizon, is to reshape evaluation rewards. Current career structures treat paper counts as a primary productivity metric (Smaldino and McElreath, 2016; Casadevall and Fang, 2014), maximising the reward r for any accepted paper. Impact-weighted or reproducibility-indexed assessment would reduce the effective r for marginal submissions while preserving incentives for high-quality work (Stephan, 2012).

Asymmetric recovery means that gradual, journal-by-journal reforms are insufficient once the system has collapsed. In Global Corruption, no individual author has an incentive to deviate back towards honesty (Scheffer et al., 2001). Recovery requires a large, coordinated intervention that simultaneously restores screening quality, raises review capacity, and rebuilds submission friction—the equivalent of a field-wide mandate rather than a journal-level policy, as illustrated by the ClinicalTrials.gov registration mandate (Moher et al., 2009).

The welfare analysis (Appendix D of the Supplementary Material) formalises the normative case. Indiscriminate submissions impose two unpriced externalities: a congestion externality that degrades review quality for all authors, and a certification externality that dilutes the precision of the certified pool for all readers (Akerlof, 1970; Ostrom, 1990). Reducing λ is a Pareto improvement; raising K_{rev} is welfare-improving when the congestion constraint binds; expanding K without reducing λ is the unique lever that unconditionally reduces welfare.

Testable predictions. The bistable structure generates two core empirical signatures. First, quality degradation should be nonlinear: metrics hold approximately steady while AI adoption rises below the collapse threshold, then deteriorate rapidly—a cliff, not a slope (Scheffer, 2009). Kobak et al. (2025) find

AI-modified text rates exceeding 40% in some biomedical sub-disciplines while remaining near zero in others, suggesting that different fields sit at different distances from their respective thresholds. Second, the gap between collapse and recovery thresholds widens as adoption increases: the intervention required to restore quality in a field that has collapsed is strictly larger than the intervention that would have prevented the collapse. Pre-AI evidence confirms this asymmetric recovery pattern: Bik et al. (2016) document that paper-mill image duplication became entrenched in specific sub-fields despite sustained editorial intervention, and Van Noorden (2023) report that annual retractions exceeded 10,000 in 2023 without receding. A direct test would compare the institutional effort required to restore quality in a collapsed field against one that never collapsed but faces equivalent submission pressure; the November 2022 release of ChatGPT provides a natural experiment for difference-in-differences identification across fields with varying AI amenability. Extended predictions, data sources, and falsification conditions are developed in Appendix E of the Supplementary Material.

Model boundaries. The model makes two structural choices that bound its applicability. First, we reduce the strategy space to two options; real authors calibrate effort continuously, and intermediate strategies can coexist in finite communities. We justify the binary reduction by an evolutionary instability argument (Weibull, 1995): intermediate submission rates are unstable under replicator dynamics and the population is driven to one extreme in the mean-field limit. In practice, a continuum of effort levels would smooth the collapse threshold into a region rather than a sharp boundary, making early warning signals harder to detect but not eliminating them. Second, we model a single journal with fixed capacity. In reality, rejected papers cascade down a prestige hierarchy: Siler et al. (2015) find that roughly 62% of papers rejected at high-impact journals were eventually published elsewhere. A multi-journal extension would predict that collapse first occurs at lower-tier venues—which typically have smaller review capacity and weaker screening—and then propagates upward. Our single-journal results are therefore conservative: the threshold at which quality degrades across the system is likely lower than the threshold for any single venue. Model extensions—multi-journal competition, heterogeneous adoption, and endogenous reviewer effort—are developed in Appendix F of the Supplementary Material.

The window for action. The window for preventive action is narrower than it appears. Because the system can remain near the honest equilibrium while structural fragility accumulates, the observable quality signal gives no reliable warning before a critical transition. Fields that invest in verification infrastructure before AI adoption deepens can preserve the integrity of certification at moderate cost. Fields that wait for a visible quality crisis will face a recovery problem requiring a larger and more disruptive coordinated intervention. The asymmetry between production and verification is not temporary: it reflects a fundamental difference in what language models can automate. Institutions designed for an era in which producing a credible manuscript was itself difficult will need deliberate redesign to remain effective in an era in which the difficulty has shifted entirely to verification.

4 Materials and Methods

Population and strategies. A population of N authors submits to a single journal over repeated periods. Each manuscript is good ($q = 1$) with probability α or bad ($q = 0$) with probability $1 - \alpha$. Under *selective submission* (OG), an author submits only when $q = 1$; under *universal submission* (AS), an author submits regardless of quality. The share of OG authors is $x \in [0, 1]$.

AI adoption channels. AI adoption $a \in [0, 1]$ affects the system through four functional forms. Submission volume: $V(x, a) = N[1 - (1 - \alpha)x] \cdot (1 + \gamma a)$, where $\gamma > 0$ is the production sensitivity. Submission cost: $c(a) = c_0(1 - \phi\, a)$, where $\phi \in (0, 1)$ governs friction erosion. False-positive rate: $\lambda(a) = \lambda_0 + \mu\, a$, where $\mu \geq 0$ governs screening degradation. Review capacity: $K_{rev}(a) = K_{rev}(0) \cdot (1 + \delta\, a)$, where $\delta \geq 0$ is the verification sensitivity. The asymmetry assumption $\gamma > \delta$ ensures that submission volume grows faster than review capacity.

Two capacity constraints. Screening intensity is $\eta(x, a) = \min(1,\ K_{rev}(a)/V(x, a))$: when submissions exceed review capacity, scrutiny per paper falls. Peer review is an imperfect filter: a good paper is falsely rejected with probability ε; a bad paper wrongly passes with probability $\lambda(a)$. When $\eta < 1$, the effective pass rates are $\pi_0 = 1 - \eta(1 - \lambda)$ (bad paper) and $\pi_1 = 1 - \eta\varepsilon$ (good paper); as $\eta \to 0$, both converge to 1 and the filter loses all discriminatory power. The peer-accepted pool is $M(x, a) = (1+\gamma a)[N\alpha\,\pi_1+N(1-\alpha)(1-x)\,\pi_0]$. The rationing probability is $\bar{\rho}(x, a) = \min(1,\ K/M(x, a))$.

Payoffs and dynamics. An OG author earns $f_{OG} = \alpha[r\bar{\rho}\,\pi_1 - c(a)]$; an AS author earns $f_{AS} = f_{OG} + (1-\alpha)\Pi_{bad}$. The replicator equation $\dot{x} = -x(1-x)(1-\alpha)\Pi_{bad}$ governs the population dynamics, where $\Pi_{bad} = r\bar{\rho}\pi_0 - c(a)$. The four long-run outcomes—Global Honesty, Global Corruption, Bistability, and Stable Coexistence—are determined by the signs of Π_{bad} at $x = 1$ and $x = 0$.

Publication quality. We define quality as the share of genuinely good papers among those accepted: $Q(x) = \alpha\,\pi_1/[\alpha\,\pi_1 + (1-\alpha)(1-x)\,\pi_0]$. Since slot rationing applies uniformly, it does not alter the good-to-bad ratio; quality is determined entirely by the composition x at the long-run outcome.

Parameter calibration. We use two representative parameter sets. Set A ($N = 1000$, $K_{rev} = 300$, $K = 400$, $r = 100$, $c_0 = 25$, $\lambda_0 = 0.1$, $\varepsilon = 0.05$) represents a typical journal prone to bistability, where the verification constraint is binding. Set B ($N = 1000$, $K_{rev} = 1200$, $K = 180$, $r = 150$, $c_0 = 30$, $\lambda_0 = 0.4$, $\varepsilon = 0.01$) represents a top-tier journal where extreme slot scarcity is the defining constraint. AI sensitivity parameters are $\gamma = 0.5$, $\delta = 0.1$, $\phi = 0.4$, $\mu = 0.2$ throughout. $\alpha = 0.2$ (20% of manuscripts are genuinely good). Full calibration rationale is in Appendix G of the Supplementary Material.

Agent-based validation. We validate the deterministic results with a stochastic agent-based model of N authors using pairwise Fermi strategy revision (Szabó and Tőke, 1998; Traulsen and Hauert, 2009). AI adoption rises along a linear ramp from $a = 0$ to $a = 0.8$ over the simulation horizon. Results across $N \in \{200, 500, 1{,}000\}$ with 100 replications each are reported in Appendices B and C of the Supplementary Material.

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